Graphical Processing Systems

Lectures and practical works

- Lectures:

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- Practical works:

  Victor Bacu, Conf. Dr. Eng.
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  Raul Gorgan, Eng.Drd., Serban Marinescu, Eng.Drd.
  Andrei Istrate, Eng., Andreea Căpriță, Eng.
  Laurentiu Grad, Eng.

- Laboratory: 105 (Dorobantilor), BT5.05 (G.Baritiu)

Activities and Knowledge assessment

1. Course
2. Practical works (Labs, Quiz, Project)

Evaluation:

50% (E), 10% (AC) - Evaluation is performed through written exam and activity at the course

40%(L) - Evaluation is performed through practical examination (laboratory, quiz, and project presentation).

Requirements to participate to exam: C≥5 and L≥5

Final mark: M=0.5*E+0.4*L+0.1*AC

Graduation requirement: M≥5, final mark

Course Syllabus

- Hidden line and surface presentation
- 3D object representation
- Polygonal objects tracing. Coordinate systems
- Local reflection model: Phong model
- Illumination models
- Shadow generation techniques
- Textures. Mapping techniques
- Global reflection models: Ray-tracing, Radiosity
- Animation techniques
- Graphics formats
- Graphical processing units
- Parallel graphics visualization
- Graphics application development environments
- Graphics libraries (OpenGL, DirectX, ...)
- Graphics languages (SVG, VRML, XML, HTML, WebGL)

Laboratory Syllabus

- Mathematics for computer graphics
- Basic structure of the Open GL application
- Graphics pipeline
- Basic vertex and fragment shaders
- Graphics primitives and transformations in OpenGL
- Graphics models, file formats
- Projections, clipping planes
- illumination models, texture mapping, and shadow computation
- Environment mapping
- Ray tracing algorithm
- Bump mapping

References

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3. Lengyel E.: "*Mathematics for 3D Game Programming and Computer Graphics*", 3rd Edition, 2012.
4. Pharr M., Humphreys G.: "*Physically-Based Rendering*", Morgan Kaufmann, 2nd edition, 2010.
5. Shreiner D., Sellers G., Kessenich J., Licea-Kane B., "OpenGL Programming Guide", Addison-Wesley.
6. Vince J.: "3-*D Computer Animation*", Addison-Wesley Publ., 1992.
7. Watt A., Policarpo F.:"*3D Games. Real-time Rendering and Software Technology*". Addison-Wesley, 2001.
8. Watt A.:"*3D Computer Graphics*". Addison-Wesley, 3rd Edition, 2013.

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9. OpenGL.ORG - The official site for OpenGL information.
10. CGAL - Computational Geometry Algorithms Library
11. Indispensable Computer Graphics Sites
12. w3schools - Data Structures and Algorithms
13. Computer Graphics Programming Resources
14. Dictionary of Algorithms and Data Structures
15. NVIDIA Technology